



SANDEEP SHARMA | 21EC39045



B.Tech.(Hons.) in ELECTRONICS AND ELECTRICAL COMMUNICATION ENGINEERING and M.Tech. in VISION AND INTELLIGENT SYSTEMS

EDUCATION

Year	Degree/Exam	Institute	CGPA/Percentage
2026	M.Tech, Dual Degree 5Y	IIT Kharagpur	8.01
2021	Class XII, CBSE	N K Bagrodia Public School, Delhi	87.8%
2019	Class X, CBSE	N K Bagrodia Public School, Delhi	91.4%

INTERNSHIPS

AI Research Intern | IBM Research, Bengaluru May 2025 - Aug 2025

(Built a 5-stage pipeline "AI4PACE - Policy Automation Compliance Engine" to Secure AI Agents).

- Created a **Policy Aware React** agent to validate the compliance ability of different LLMs, further added roles to make it **RBAC** policies.
- Experimented with various **DSL** (XACML, OPA) to write formal policy, later created a Rego code generator with **Agentic-RAG** capabilities.
- Implemented a **Policy-Advisor** agent to help in writing the policy alongwith **Validator** agent for testing them through synthetic test cases.
- Filed an **Invention** disclosure (P202502638) "Advisor for Dynamic Policy Generation & Refinement in the Agentic Applications".
- Inventors: Akshar Kaul, Venkatraman Ramakrishna, Sandeep Sharma, Sandeep N, Abhishek Singh, Vivek Sharma, Vinayaka Pandit

Causal AI Research Intern | Ericsson Research, Bengaluru May 2024 - Jul 2024

(Developed and Benchmarked Advanced Causal Discovery Methods for Reinforcement Learning Environments).

- Benchmarked **8+** causal discovery algorithms on RL dataset, achieving **95%** TPR with DAG-GNN, reduced **SHD** by **60%** on Asia dataset.
- Enhanced **DAG-GNN**, **GAE** architectures with prior knowledge loss injection, improving FDR from **0.4** to **0.16** and increasing TPR by **25%**.
- Used component-wise neural (**cMLP**, **cLSTM**) for non-linear Granger causality analysis on **100K+** time-series data points from RL **buffer**.
- Validated causal learning in RL env. to compare **random action** causal graphs to trained ones (**DQN/SAC**), eliminating policy training.

AI & LLM Developer Intern | Tecnod8.ai, Mandi Aug 2024 - Apr 2025

(Customized open source LLMs for industrial knowledge, automation and built agents on top of it).

- Finetuned **Bge-base** embedding model using **MNR** loss and later added a reranking stage for improving hitting rate of RAG by **7.42%**.
- Applied **Width-wise** structured pruning, to make an efficient version of **Llama-3.1-8B**. Further used **CPT** to infuse industrial knowledge.
- Distilled knowledge from mistral-nemo, llama-70b by combining **KL divergence** loss, surpassing gpt-4o mini by **13%** on complex query.
- Used **Low rank adapters** to finetune **Codegemma-7B** for **PLC** and **CNC** code generation. It Improved average prometheus score by **27%**.
- Built a **Multimodal** model by merging **Vision encoder (Qwen2.5-VL)** with decoder (Llama) to translate 2D sketches into executable flows.

Causal Reasoning & Agentic AI Intern | Riskspan Technology Services, Chennai

(Agent-driven Causal AI that constructs and reasons over a financial knowledge graph to do root cause analysis).

- Built Agents to parse unstructured financial (**10K**) reports, systematically identifying and structuring key **entities**, **facts**, and **relationships**.
- Constructed a knowledge graph by populating a **Neo4j** database, and integrated an NL to **Cypher**, **SQL** module to query complex events.
- Used **causal** inference based principles from "**Ladder of Causation**" to validate factuality and determine root causes of the user query.

Data Science Intern | Western Digital Corporation, Bengaluru

(Developed "Symphony" Multi-Agent Architecture Combining LLMs with ML/DL for Enterprise Data Analysis).

- Architected multi-agent system integrating for 45 tables, bypassing token limits and reducing LLM costs by through data preprocessing.
- Implemented **Forest of Thoughts (FoT)** reasoning framework achieving 85% query accuracy and 95% success rate via 5-module pipeline: Question Reconstruction, Validation, Auto Prompt Construction, SQL Generation, and Self-Refining Learning.
- Built automated **SQL Agent** with dynamic query generation supporting ranking, what-if scenarios eliminating manual query.
- Automated report generation for manufacturing metrics (Switch%, cycle time, WIP flow), for real-time monitoring and decision making

PROJECTS

Adversarial Framework for Multi-Scale Anomaly Detection | Dr Ye Zhu | Deakin University Apr 2024 - Aug 2024

(Anomaly detection using feature pyramids, adversarial CNN training and student-teacher approach).

- Designed multi-scale anomaly detection model with student-teacher and feature pyramids, achieving **AUC** of **0.98+** on MVTec-AD dataset.
- Leveraged a **ResNeXt-50** backbone and adversarial training to enhance feature extraction, improving detection **precision** by **15%**.
- Validated model performance at both image and pixel levels, securing **ROC** scores over **0.98** for all test categories.

Drone Images Acacia Detection with GANs | Dr. Daphne Teck Ching lai | University of Brunei. Jan 2024 - Mar 2024

(Anomaly detection using feature pyramids, adversarial CNN training and student-teacher approach).

- Automated acacia tree detection from high-resolution imagery, using conditional **GANs** integrated with attention-based **U-Net** model.
- Implemented multi-scale patch extraction with precise segmentation masks and used **HSI** features for enhanced classification.
- Integrated C-GAN **discriminator** with residual attention U-Net, achieving **95.73% mIoU score**, demonstrating best performance.

SKILLS & COURSEWORKS

SKILLS:- Langgraph | Huggingface | PyTorch | Torchvision | OpenCV | Streamlit | Python | C++ | MATLAB | MySQL | LaTeX | Git

COURSES:- Pattern Recognition | Deep Learning* | Image & Video Processing* | NLP | Interpretable Machine Learning | Vision & Visualization | Data analytics | Probability & Statistics | Algorithms | ML & DL specialization (Deeplearning.ai)** | Machine Learning A-Z